

THE DENARIO PROJECT: DEEP KNOWLEDGE AGENTS FOR SCIENTIFIC DISCOVERY



SCAN ME



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[HTTPS://GITHUB.COM/ASTROPILOT-AI/DENARIO](https://github.com/ASTROPILOT-AI/DENARIO)



UNIVERSITAT DE
BARCELONA



Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA



MINISTERIO
DE CIENCIA, INNOVACIÓN
Y UNIVERSIDADES



AGENCIA
ESTATAL DE
INVESTIGACIÓN



Villanueva-Domingo
(Computer Vision
Center)



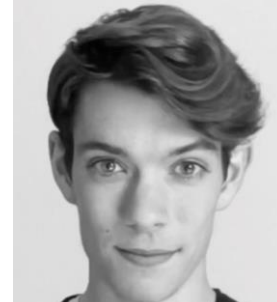
Bolliet
(Cambridge)



Villaescusa-Navarro
(Flatiron)



Bayer
(Flatiron/Princeton)



Bermejo
(DIPC/Flatiron/Google)



Bilodeau
(Virginia)



Hahn
(Arizona/Austin)



Jimenez
(Barcelona)



Zou
(SISSA)



Lee
(Flatiron/Princeton)



Meyer
(LMU)



Protopapas
(Harvard)



Roy
(Virginia)



Tarancon-Alvarez
(Barcelona)



Wang
(DeepMind)



Xu
(Cambridge)



Zhu
(Flatiron)



Spergel
(Simons)



Anand Ojha
(Flatiron)



França
(Harvard)



Zhou
(UTokyo)



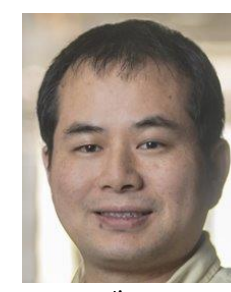
Wang
(UChicago)



Cardenas-Ramirez
(Cornell/MIT/Harvard)



Lerario
(SISSA)



Jiang
(Flatiron)



Acquah
(Oxford)



Amancharla
(Infosys)



Wadekar
(John Hopkins)



Barzilay-Siegal
(Tel-Aviv)



Yovel
(Tel-Aviv)



Viel
(SISSA)



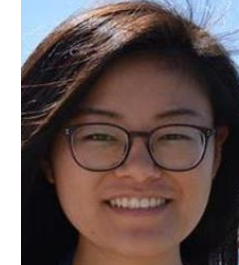
Cranmer
(Cambridge)



Zubeldia
(Cambridge)



Mamun
(Cornell)



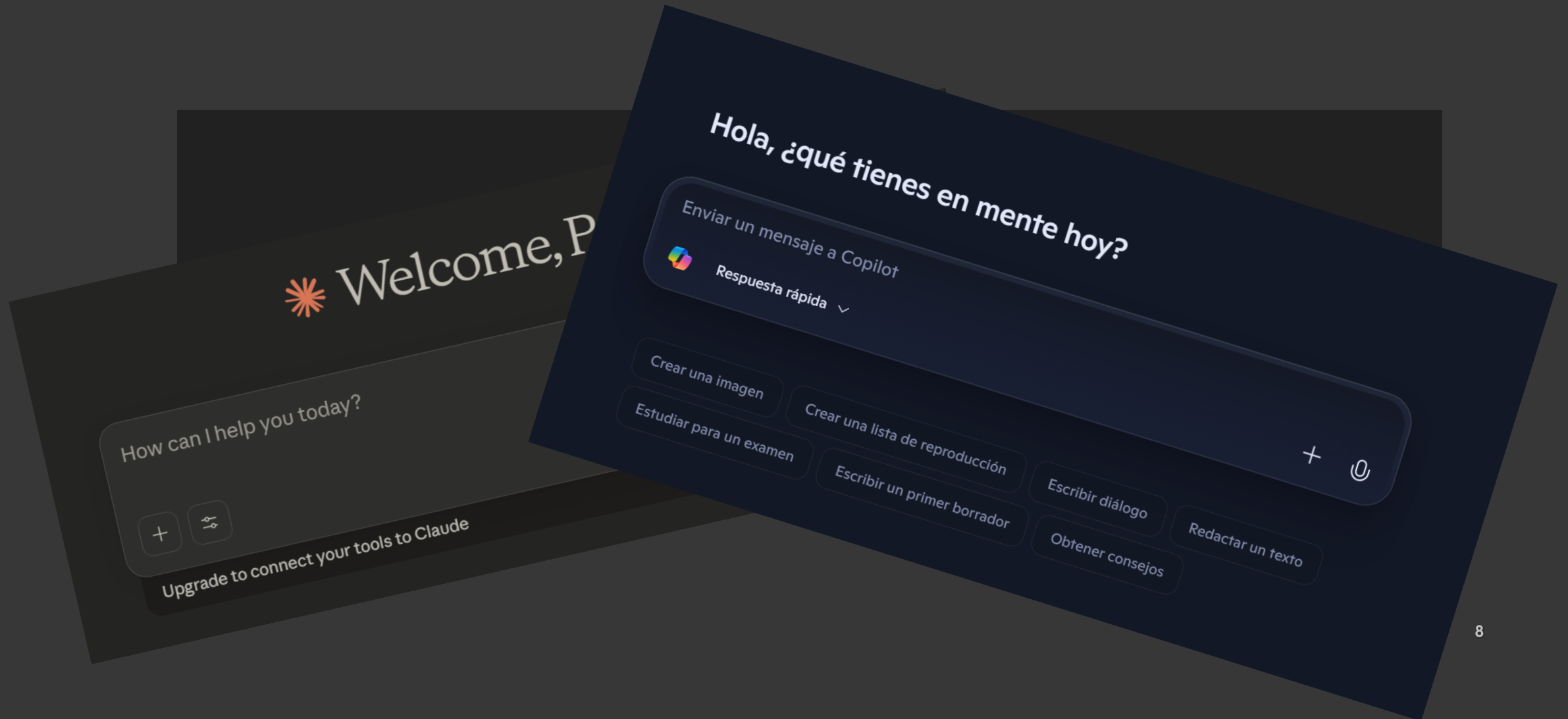
Yue
(Cornell)



Tiwari
(Infosys)

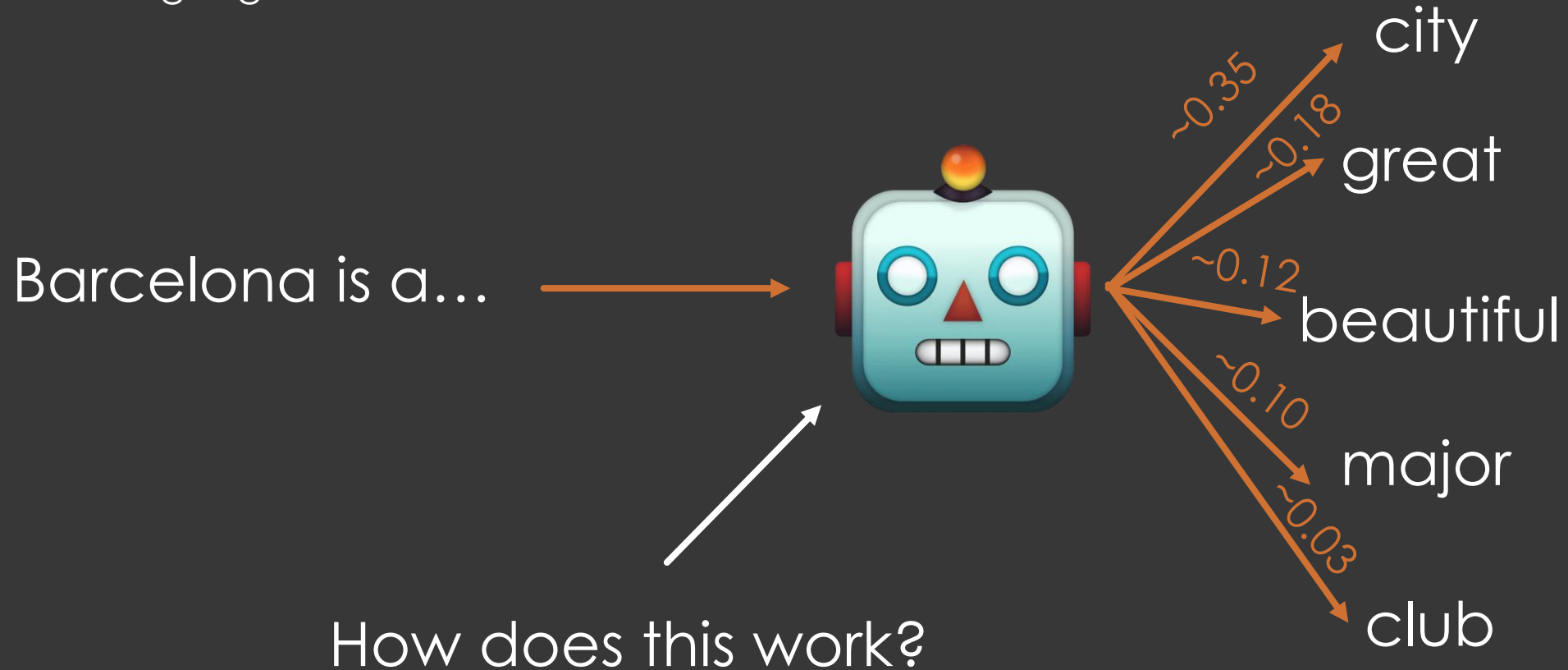


LARGE LANGUAGE MODELS

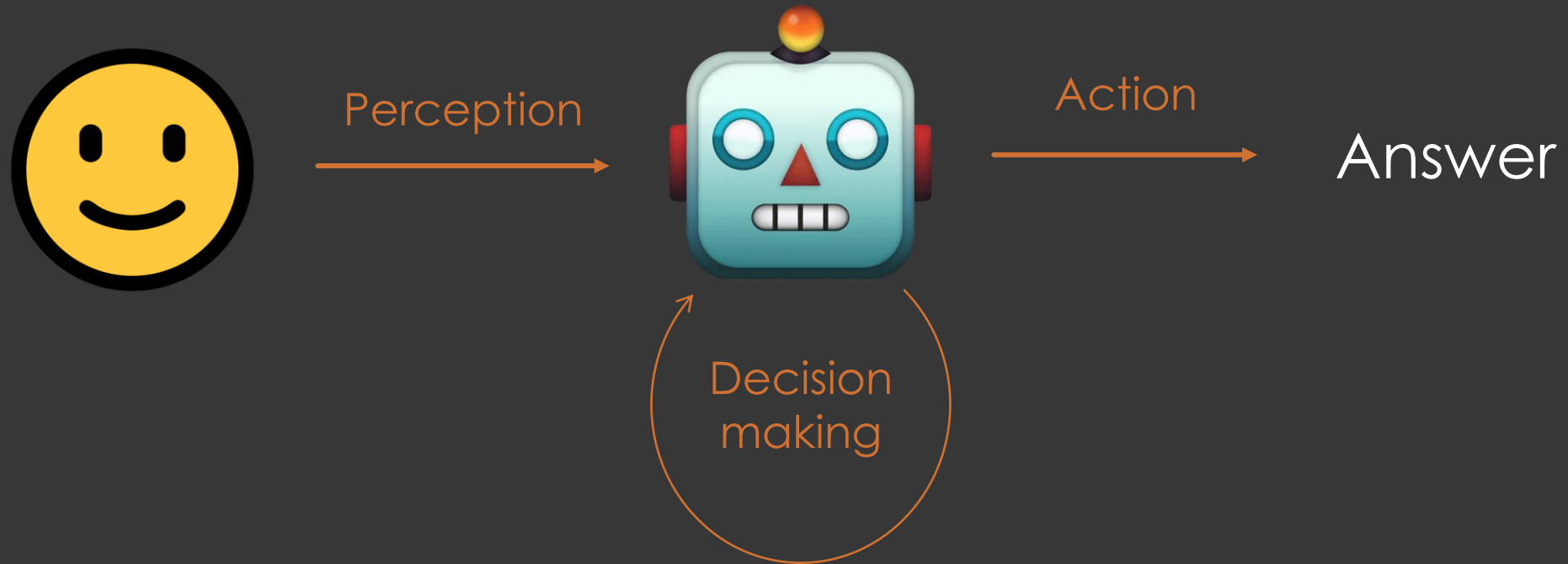


LLM

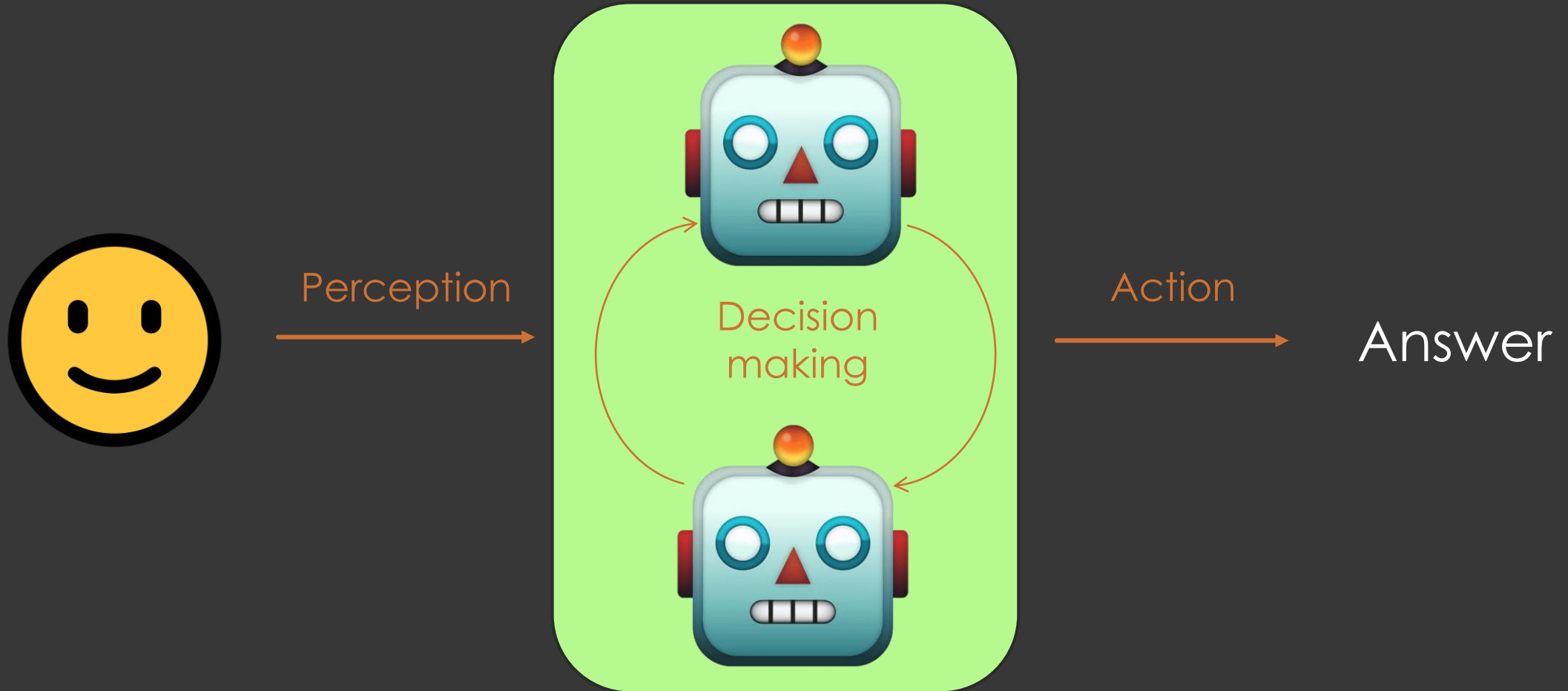
- Type of neural network trained on massive text datasets to predict and generate natural language.



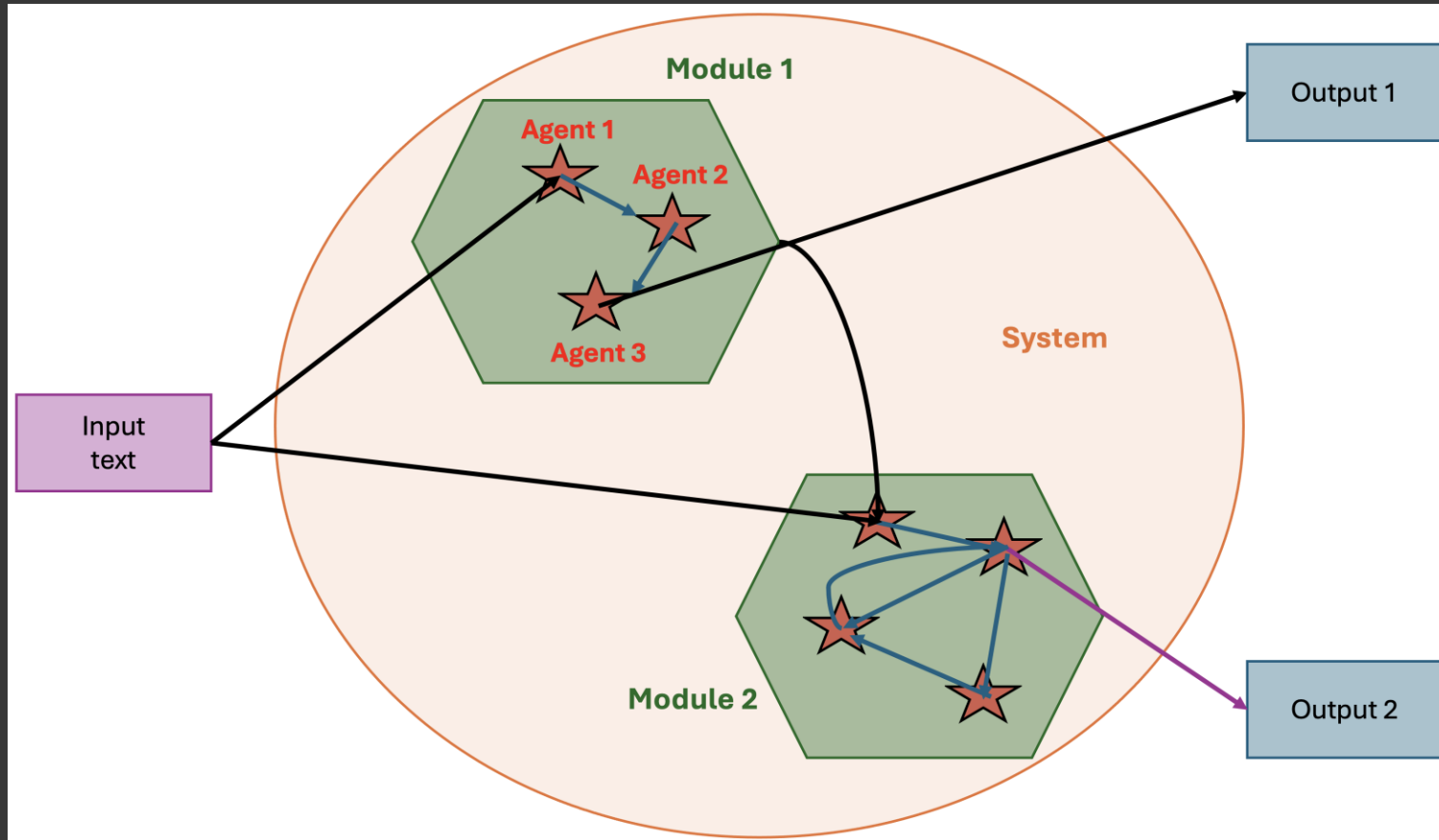
AI agent system



AI multi-agent system



AI multi-agent system



CAN WE DO SCIENCE WITH
AGENTS?

The Denario Project: Modular Automation of Scientific Research with Multi-Agent Systems

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Bonny Y. Wang³⁵, Licong Xu^{36,4}, Yossi Yovel^{8,37}, Shuwen Yue¹³, Wen-Han Zhou³⁸,
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Denario: Modular Automation of Research

<https://github.com/AstroPilot-AI/Denario>

- AI multi-agent system designed to be a scientific research assistant
- Generate ideas, check the literature, develop research plans, write and execute code, make plots, and write a scientific paper
- Modular system: multi agent orchestration with AG2 and LangGraph



<https://ag2.ai/>

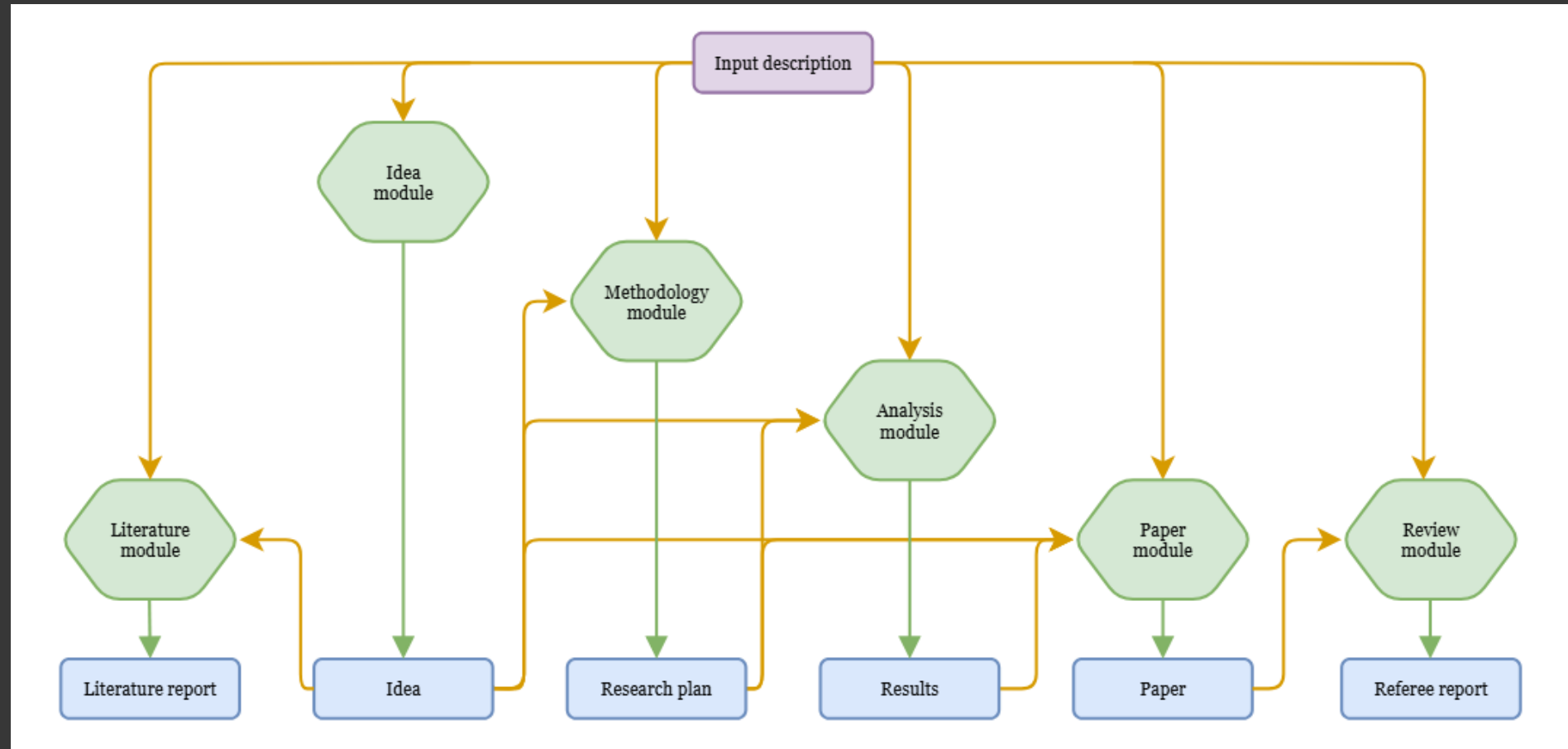


- Plan and Control is done with cmbagent

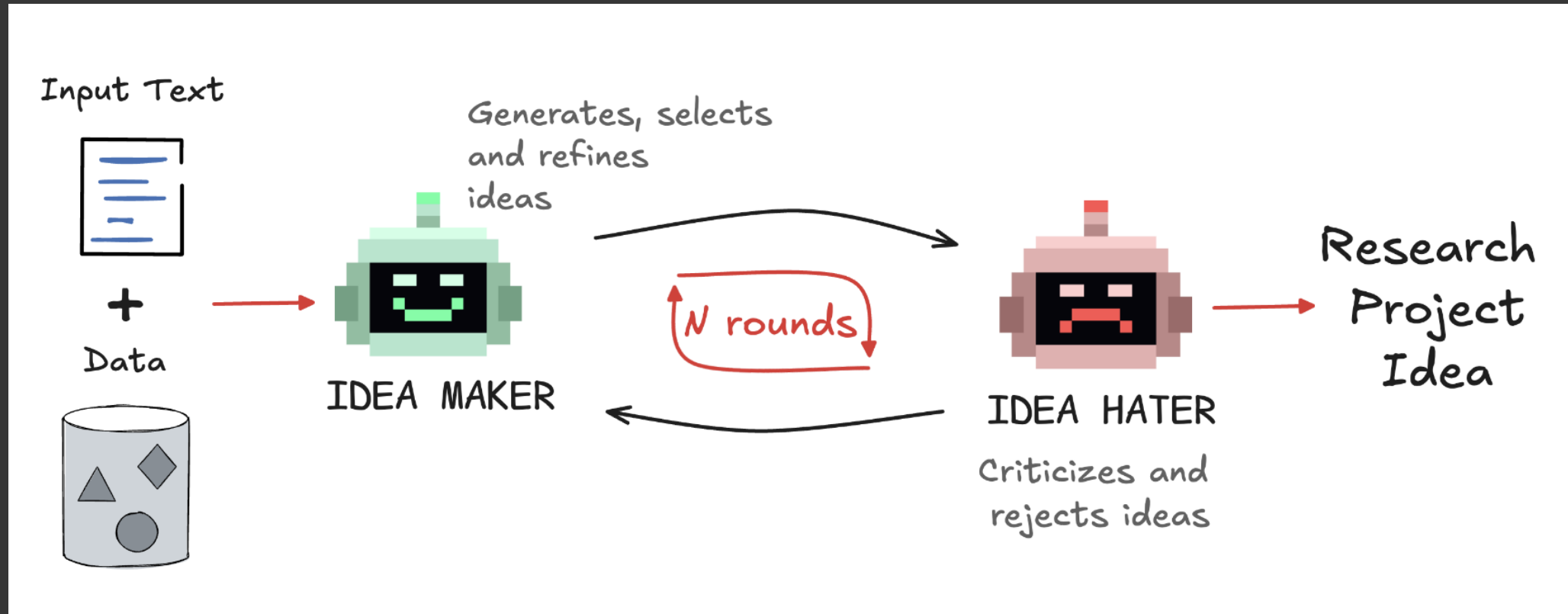


<https://github.com/CMBAgents/cmbagent>

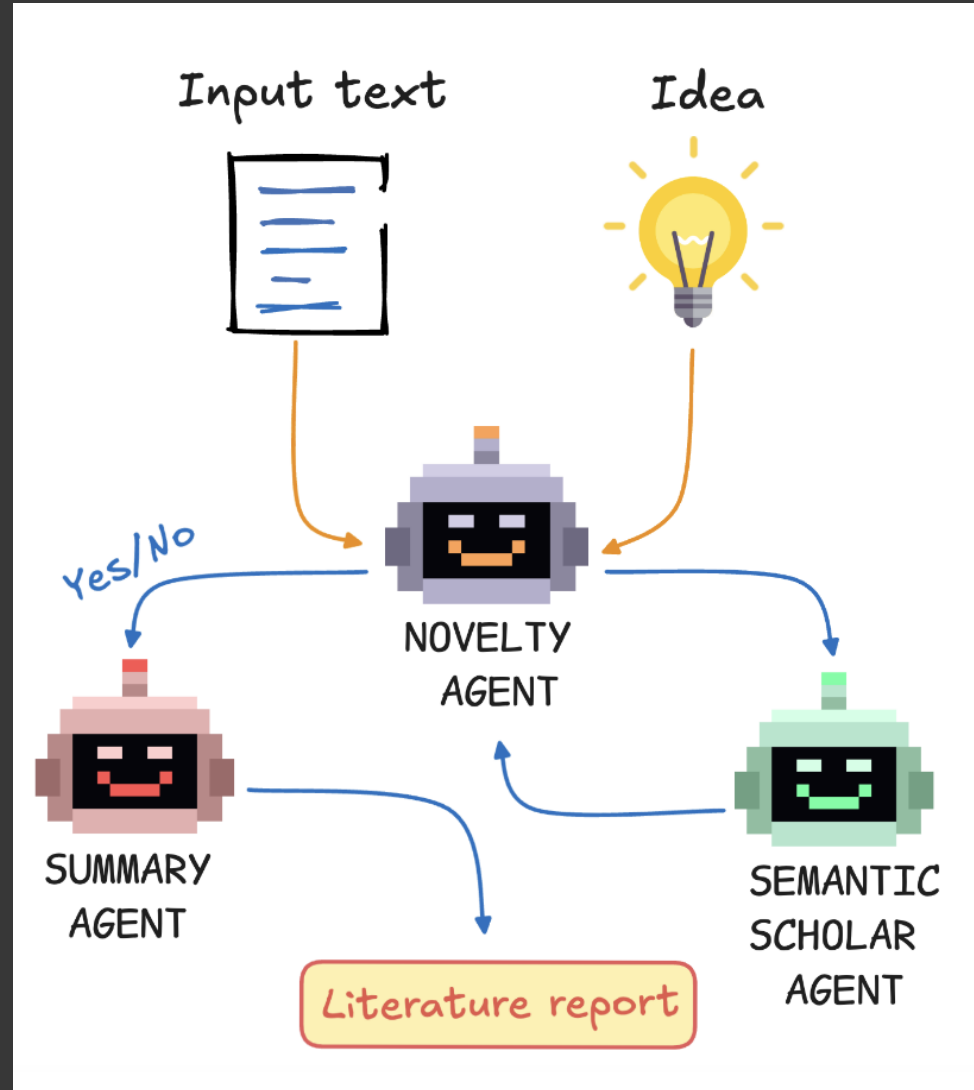
Denario: architecture



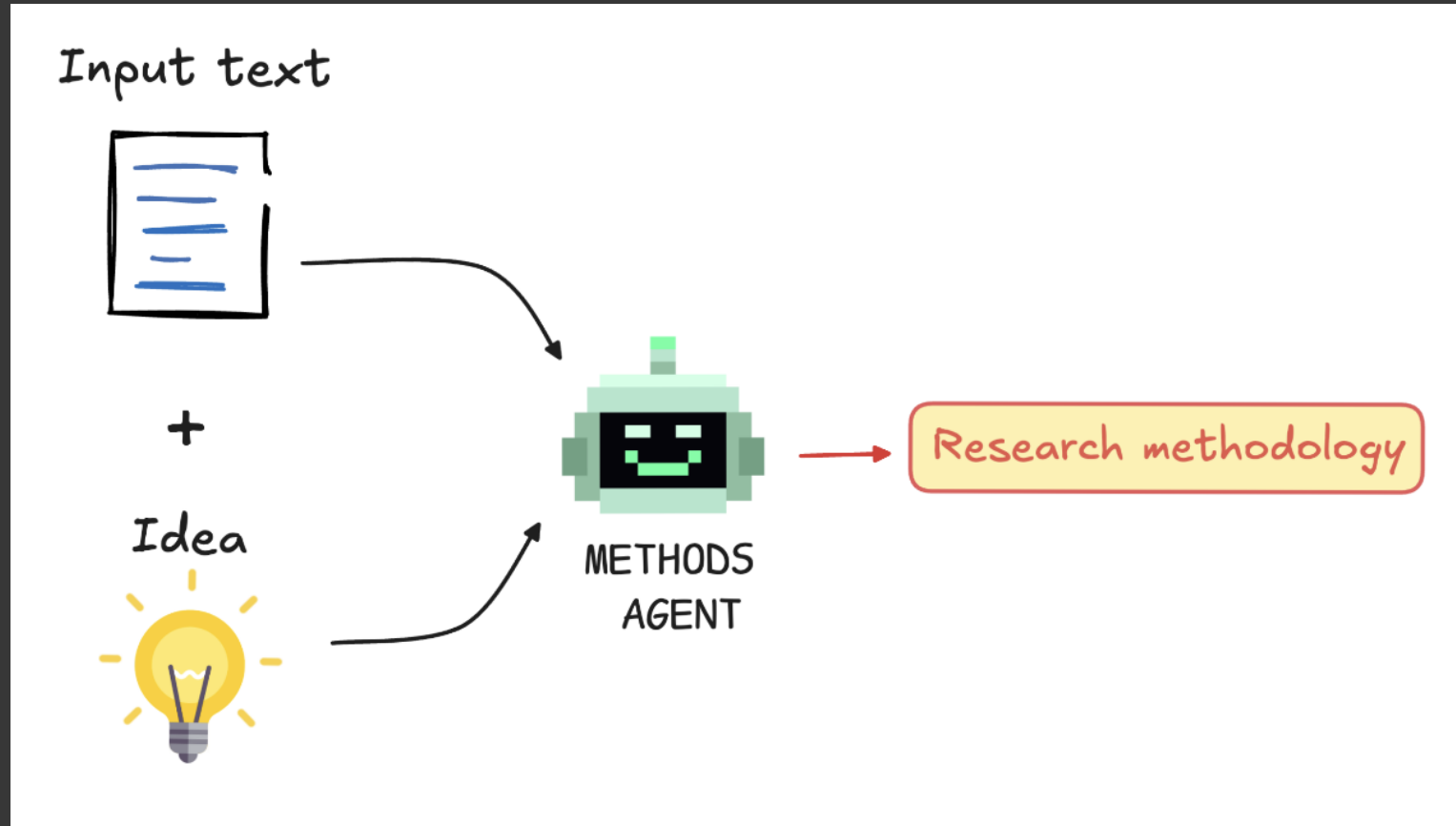
Denario: idea module



Denario: literature module

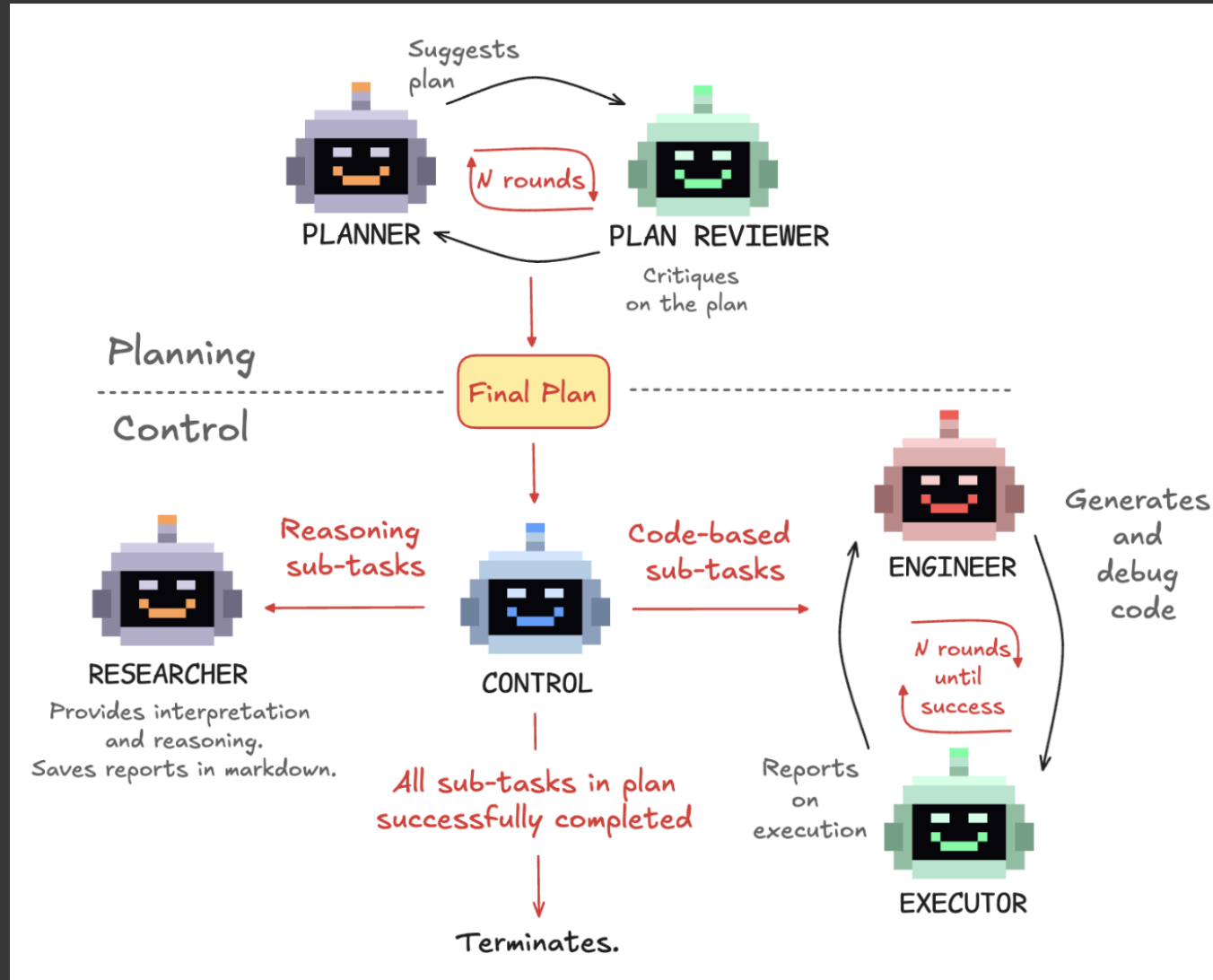


Denario: methodology module

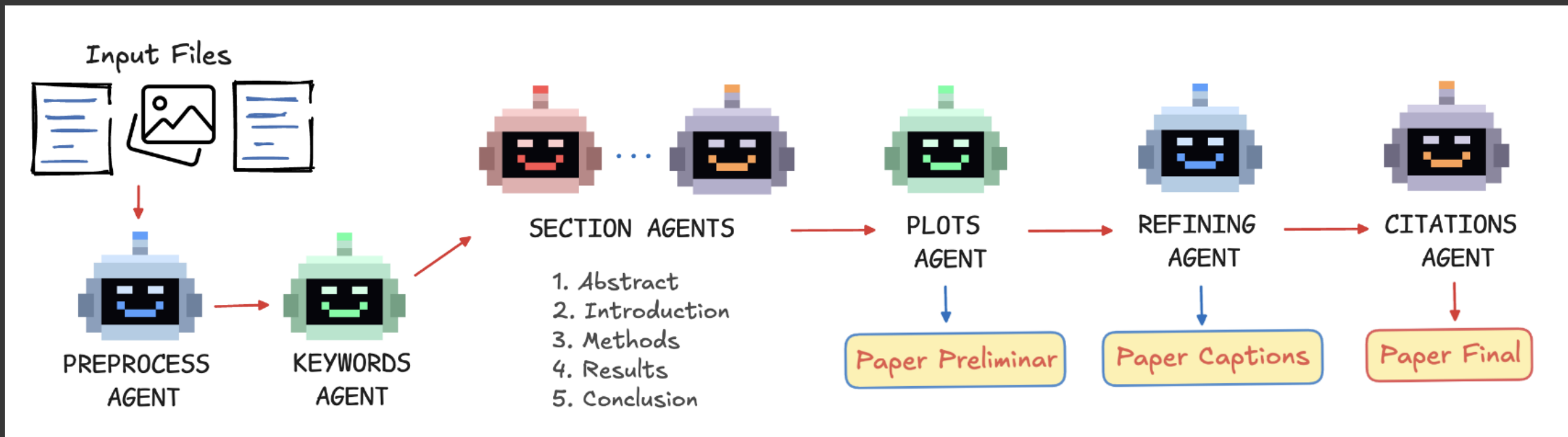


Denario: analysis module

[Xu et al. 2025]



Denario: paper module



Denario: conclusions

- Given an input text, DENARIO can generate ideas, develop research plans, Write and execute code, make plots, and summarize results and write a scientific paper
- Validation is crucial. Mimicking a scientific paper structure doesn't mean it is valuable
- Advantages: replicability and modular structure. It can be used as a tool to speed up tedious processes
- Denario is publicly available, although OpenAI API key is mandatory to run the whole pipeline

How does an LLM work?

Input text → Tokenization → Embeddings → Transformers → Prediction

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- The text that we want to be continued

"The cat that chased the mouse was hungry"

How does an LLM work?

Input text → Tokenization → Embeddings → Transformers → Prediction

- Split the text into different words
- Each word is then mapped to a number depending on the algorithm used

"The cat that chased the mouse was hungry"

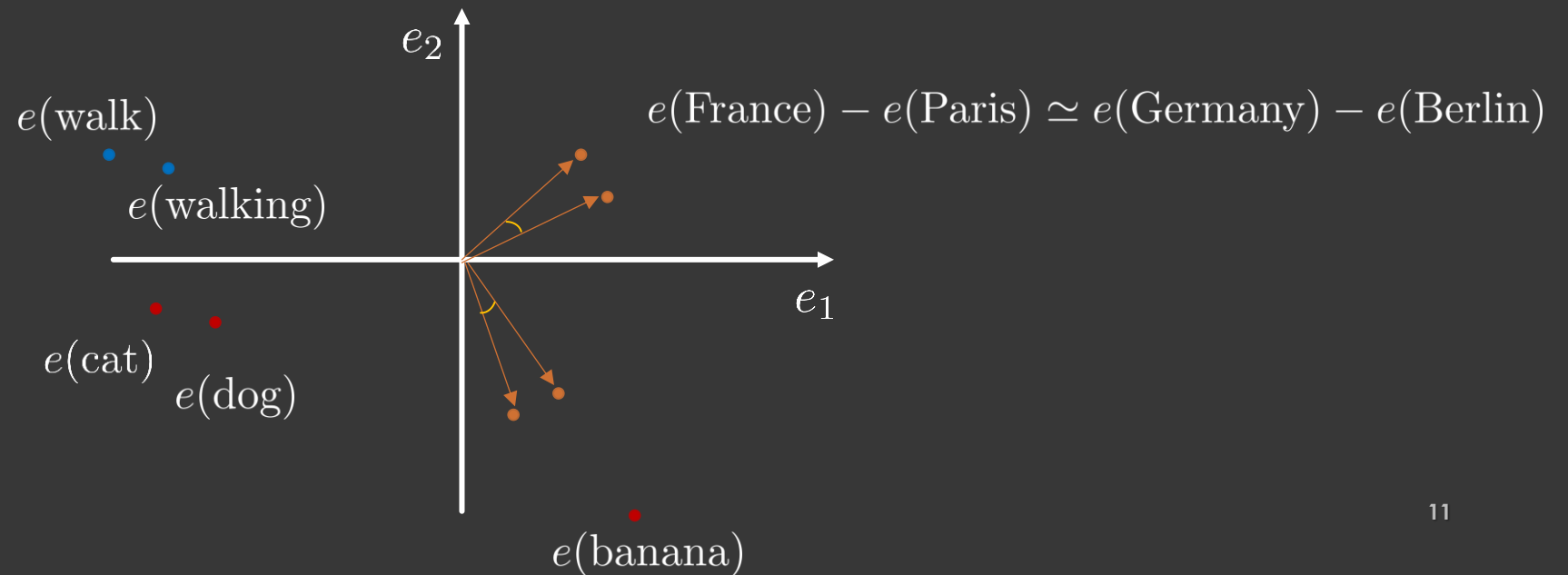
BERT's WordPiece tokenizer

[101, 1996, 4937, 2008, 4025, 1996, 4301, 2001, 5823, 102]

How does an LLM work?

Input text $\longrightarrow$ Tokenization $\longrightarrow$ Embeddings $\longrightarrow$ Transformers $\longrightarrow$ Prediction

- Each token is mapped to a vector that lives in a high dimensional space
- The embeddings capture semantic similarity (position encoding)



How does an LLM work?

Input text → Tokenization → Embeddings → Transformers → Prediction

- Reason why modern LLMs work [Vaswani et al. 2017]
- Each transformer has: self attention layers + feed forward layers

Self attention

- Each token looks to the rest
- How much each token should “attend”
the rest is quantified by attention weights

"The cat that chased the mouse *was* hungry"



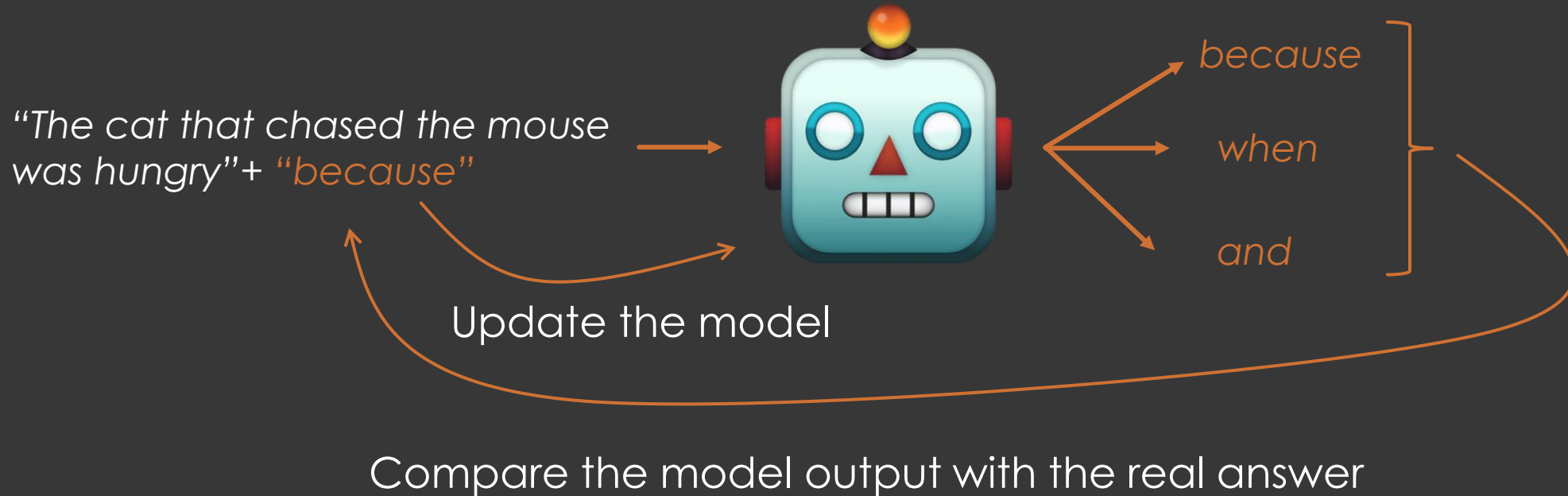
Feed forward layers

- Tokens pass through several fully-connected neural networks
- Multihead structure: each of them with their own attention

Transformers are stacked in consecutive layers

How does an LLM work?

Input text → Tokenization → Embeddings → Transformers → Prediction



AI multi-agent system

[Xiao et al. 2025]

AI multi-agent system to replicate real-world trading strategies

